BIO411/BIO629 End semester Exam (2025) IISER Mohali

Date: *21/11/2025* **Duration:** *90 min*

Maximum marks: *20*

Instructions:

a. *Answer questions briefly in section. Include equations, wherever applicable with all terms defined appropriately.*
b. *All questions in section A are compulsory. Answer any 2 questions from Section B. Each question carries 5 marks.*

Section A

1. What are sequence domains? Write Hidden Markov Model architecture to define sequence family profile. **(2 marks)**

   Sequence domains are evolutionary conserved regions within protein sequences and is usually associated with protein function.

2. How to determine secondary structure of protein with its contact map? **(2 marks)**

   Contact map is 2D matrix representation of spatially contacting residues. In a contact map generated using CA contact distance of 7Å, lines (dense bands of contact) parallel and immediately adjacent (close) to the main diagonal represent helix, lines parallel to main diagonal with offset from it is by parallel strand and lines perpendicular to diagonal are antiparallel strand.

3. Protein tertiary structures often reveals significant structural similarity despite having very low sequence identity. What main conclusions can be drawn from this observation, and discuss its major impacts on the field of bioinformatics? **(2 marks)**

   The main conclusions drawn from the statement is that structures are more conserved that sequences and structure plays essential role in determining its function such as functional sites depend on spatial arrangement of residues. Additionally, structure conservation indicates possible arrangements of secondary structure is limited either due to physiochemical restraints or evolutionary selection constraints. Many impacts resulting due to this observation: a. Protein structure prediction as number of folds is limited b. Classification of folds to annotate protein structure and function, c. Structural similarity methods for function annotation, d. Molecular evolution

4. Describe and use schematic diagram to show dihedral angles *ϕ, ψ, and ω* for GLY dipeptide. Discuss briefly basis used in defining allowed, partially allowed and disallowed regions in Ramachandran map. Which L-amino acids will have similar distribution in R-map. Discuss reasons of amino acids showing variation **(4 marks)**

   Allowed: Conformations are completely/fully allowed, if distances between all pairs of non-bonded atoms are greater than normal limit (distances are larger than sum of van der waals radii considering hard sphere model)

Partially allowed: Conformations are partially allowed, if distances between one or more pairs of non-bonded atoms are less than normal limit but are greater than outer limit (distances are larger than sum of van der waals radii considering soft sphere model)
Disallowed: Conformations are disallowed, if distances between one pair of non-bonded atoms are less than outer limit.
All amino acids except Glycine and Proline have similar distribution. Glycine due to absence of Cbeta has larger allowed region. Proline has restricted phi angle in R-map because cyclization of side chain on backbone restricts dihedral angle rotation.

Section B

5. α-synuclein consists of intrinsically disordered region (IDR) facilitating its binding to multiple proteins. Protein Data Bank (PDB) contains three distinct structures of α-synuclein bound to different interacting partners and one structure not bound to any partner. How can the specific interaction regions on α-synuclein be identified in complexes? How can we determine if these interactions occur through an induced-fit mechanism? Suggest appropriate measure of evaluation for the latter.

   Specific interaction between synuclein and its partner can be deduced using: change is relative SASA (RSA) for each residue upon complex formation; atomic contact between two proteins.
   Perform sequence based structure alignment to make inference whether there is conformational change upon binding of synuclein partner. The reference state is apo- (partner unbound) form. If there is conformational change on binding one can suggest there is induced-fit mechanism. The metric for conformational change is RMSD.

6. A protein X was identified from gut metagenome sequencing project. It showed no sequence homology to a protein of known function. Suggest appropriate *in silico* approaches, which can be used for annotating protein X. Suggest a tool/method which can be used for such analyses.

   Details of non-homology based method should be discussed. Use of STRINGdb. If you want to use structure for function determination than one should mention that structure can be predicted using any prediction method followed by structure based methods for function prediction as: structural similarity, structural motif determination, functional site similarity.

7. A skin metagenome sequencing project recovered 10 draft genome sequences. A particular protein is present in all 10 genomes is also shown as conserved among archaea. Propose approaches that can be used to investigate the evolutionary relationships of these genomes with archaea. For any one of approach, outline the key analytical steps you would follow to assess their evolutionary relatedness. Define horizontal (lateral) gene transfer.

   Describe various steps of any phylogenetic tree reconstruction method to analyze evolutionary relationship between 10 genomes with archaea. Searches of homologs in other *bacteria or other eukaryote should be included. MSA followed by distance among sequences (1-sID) or use of appropriate JTT/Dayhoff matrix (for distance based method). In the final step after tree reconstruction, it should be discussed whether archaeal sequence is within the metagenome clade to make suitable conclusion. Horizontal gene transfer can be described as a process wherein an organism transfers genetic material (genes) to another organism that is not its offspring. For instance antibiotic resistance is result of HGT.*

8. What is affine gap penalty and its significance in pairwise sequence alignment? Suggest conditions when global sequence alignment will be useful. Perform global sequence alignment of sequence HEAGAW with PEAHWAW (Match: +3, Mismatch: -1, gap: -2). Report Score, sequence identity and possible alternate alignments.
Affine gap penalty results in more biological relevant alignments because the gaps (deletion) events are not frequent in consecutive positions. Global sequence alignments are useful when sequences are of similar length, and it is required to compare two sequences for their full length to infer function, orthologous or evolutionary relationships.

	-	**P**	**E**	**A**	**H**	**W**	**A**	**W**
-	0	-2	-4	-6	-8	-10	-12	-14
H	-2	-1 (D)	-3 (DL)	-5 (DL)	-3 (D)	-5 (L)	-7 (L)	-9 (L)
E	-4	-3 (DU)	2 (D)	0 (L)	-2 (L)	-4 (DL)	-6 (DL)	-8 (DL)
A	-6	-5 (DU)	0 (U)	5 (D)	3 (L)	1 (L)	-1 (L)	-3 (L)
G	-8	-7 (DU)	-2 (U)	3 (U)	4 (D)	2 (DL)	0 (DL)	-2 (DL)
A	-10	-9 (DU)	-4 (U)	1 (U)	2 (U)	3 (D)	5 (D)	3 (L)
W	-12	-11 (DU)	-6 (U)	-1 (U)	0 (U)	5 (U)	3 (U)	8 (D)

Score : 8

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S1: PEAHWAW
S2: HEA-GAW

S1: PEAHWAW
S2: HEAG-AW

sID: (4/7)*100 = 57.14%
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